MCO-1D

Software Reference — Rev 23

Ground station command reference for commissioning engineers. For pilot operation see the Operator Manual v23.

Document	MCO-1D_Software_Reference_v23
Revision	23 — August 2026
New in v23	Reference bugs (vsi_ref, hdg_ref, alt_ref) · precision_status · param_set · stroke smoothing · climb/speed/manoeuvre
Audience	Commissioning engineers

Architecture and Control Loop

50 Hz main loop: pilot_intent_tick → CIC (climb) → MIC (aerobatics) → wing_guard_tick → stroke_smoothing_tick → ODrive CAN output. Discovery at 1 Hz during stable cruise. Codebase: 70,356 lines. 64 hypotheses. 43 SG tasks. 15 foci. 324 GS commands.

Arm / Status

`arm`
Arm the aircraft.

`disarm`
Disarm.

`status`
Full system status summary.

`fault_summary`
Read and decode active faults.

`soft_stop_status`
All axis status.

`pilot_intent_status`
Current CIC/MIC mode, blend, VIFF, guard state.

Reference Bugs — NEW v23

- `vsi_ref [fpm]`
  Set VSI reference bug in ft/min. Always displayed on VSI gauge. Positive = climb, negative = descent, 0 = level. Range: -2000 to +2000 ft/min. Does NOT make the aircraft hold this rate. Display only. AI: "target 100 ft/min down" / "descent rate 500" / "level flight"
- `vsi_ref_clear`
  Remove VSI reference (return to zero).
- `hdg_ref [0-360]`
  Set heading bug in degrees magnetic. Displayed on directional indicator. Status strip shows bug value, current heading, and error. Does NOT make the aircraft fly this heading. AI: "set heading bug to 090" / "fly heading 180" / "bug heading north"
- `hdg_ref_clear`
  Remove heading bug from directional indicator.
- `alt_ref [altitude] [ft|m]`
  Set altitude bug. Default unit: feet. Displayed on altimeter. Does NOT make the aircraft hold this altitude. AI: "altitude bug 1000 feet" / "target 300 metres AGL"
- `alt_ref_clear`
  Remove altitude bug from altimeter.
- `precision_status`
  All three reference bugs, current values, and errors. Also reports trim quality (VSI / IAS / roll RMS) if controls have been neutral in calm air. AI: "precision status" / "VSI error" / "how accurate is the altitude hold"

Precision / Stroke Smoothing — NEW v23

- `stroke_smoothing_tick [internal]`
  5-beat rolling average for VSI, IAS, roll. Runs every tick. Removes 3.74 Hz stroke pulsation from instrument display. Telemetry keys: vsi_smooth_fpm, ias_smooth_kt, roll_smooth_deg, vsi_ref_fpm, vsi_ref_err_fpm, hdg_ref_deg, hdg_ref_err_deg, alt_ref_m, alt_ref_ft, alt_ref_err_ft.
- `param_set [key] [value]`
  Generic parameter setter. Use specific commands where available. Example: param_set cascade_phase_deg 30.0

Diagnostics — NEW v23

- `speed_performance_status`
  Speed regime boundaries, Strouhal numbers at max capability, active speed foci and hypotheses.
- `climb_performance_status`
  Both climb optima (ROC and angle), target speeds, configurations, ALSS curriculum coverage.
- `climb_intent_status`
  CIC diagnostic: current mode, blend fraction, classification thresholds. Not surfaced to pilot.
- `manoeuvre_intent_status`
  MIC diagnostic: current manoeuvre, urgency, per-manoeuvre dwell/blend. Not surfaced to pilot.

Wing Coordination

`wc_set_fuselage_width [mm]`
Set measured spar-to-spar fuselage width. REQUIRED before TC-01.

`wc_set_tpu_start [mm]`
Set measured TPU winglet boundary from root pivot. REQUIRED before TC-01.

`wc_geometry_report`
Full two-zone geometry report. Confirm 96 deg contact in TPU zone.

`wc_fuselage_guidance`
Design guidance for fuselage width selection.

`clap_enable [dorsal|ventral]`
Open clap gate. Validated wing contact in TPU zone at stroke reversal.

`clap_disable`
Close clap gate.

`clap_status`
Two-zone amplitude limits, tip separations, clap count.

`wing_guard_status`
Recent guard interventions.

Arm / Hold / Hover

`attitude_hold_status`
PID gains, trim state, attitude errors.

`hover_mode_engage`
Engage hover mode.

`hover_mode_disengage`
Disengage hover mode.

`position_hold_engage`
Engage GPS position hold.

`position_hold_disengage`
Release position hold.

`vsi_hold [fpm]`
Alias for vsi_ref — sets VSI reference bug only. Does NOT engage autopilot hold.

`track_hold`
Display reference: captures current heading for display. Does NOT make the aircraft hold the heading.

Learning / Focus

`focus [name]`
Set mission focus. See focus list below.

`focus_clear`
: Clear active focus.

`disc_registry`
: List all 64 hypotheses.

`disc_mode_focus_list`
: List all 15 foci.

`discovery_master_on`
: Enable discovery during flight.

`discovery_master_off`
: Disable discovery.

`freeze_optimizers`
: Freeze all learning.

`restore_optimizers`
: Resume learning.

`mechanism_weight_report`
: Show learned mechanism weights.

`mechanism_weight_reset`
: Reset all weights to equal.

`persist_checkpoint`
: Save learning to disk immediately.

Focus names (15 total, v23)

FOCUS	DESCRIPTION
`gust_tolerance`	Wind and turbulence stability
`position_hold`	GPS position hold accuracy
`smooth_flight`	Minimum vibration / camera payload
`crosswind_landing`	Crosswind approach and landing
`endurance`	Minimum power consumption
`sprint`	High-speed sprint with safety margin
`sar_hover`	SAR hover and confined-space ops [persistent]
`storm_resilience`	Maximum dihedral, load shedding [persistent]
`aerobatics`	Roll / pitch / yaw authority and precision
`kestrel_hover`	Thermal soaring mode [persistent]
`top_speed`	Maximum speed pursuit (28 m/s Vne)
`fast_cruise`	Efficient high-speed transit
`best_climb_rate`	Maximum rate of climb at Vy
`best_climb_angle`	Maximum climb angle at Vx
`precision`	Trim quality — minimum hands-off variation

ALSS / Ceiling

`alss_set_fence [lat] [lon] [radius_m] [height_m]`
Set practice fence.

`alss_start`
Start ALSS programme.

`alss_abort`
Abort current sortie.

`alss_status`
Sortie number, class, segment, scores.

`ceiling_explore_arm`
Step 1: checklist + 60s arm window.

`ceiling_explore_confirm`
Step 2: begin probe (within 60s).

`ceiling_explore_abort`
Descend immediately.

`ceiling_explore_status`
Altitude, saturation margins, LEV risk.

Spatial Awareness

`spatial_awareness_enable`
Enable obstacle sensing ring.

`spatial_awareness_disable`
Disable.

`obstacle_status`
Live obstacle map from six ToF sensors.

`gap_set_width [m]`
>0.55m: flapping 0.20-0.55m: sideways 0.12-0.20m: ballistic.

`gap_enter`
Begin gap negotiation passage.

`gap_abort`
Abort passage.

`canopy_approach [lat] [lon] [alt_m]`
Canopy approach to target.

`perch_land`
Vertical descent to surface.

`confined_exit [height_m]`
Pop-up vertical exit from confined space.

Knowledge / Test Cards

`knowledge_export [path]`
Export knowledge base to JSON. Default: /home/pi/knowledge/export_TIMESTAMP.json

`knowledge_export_usb`
Export to USB if mounted.

`knowledge_import_usb`
Import from USB.

`knowledge_boot_check`
: Verify knowledge directory on Pi.

`tc01_start / tc01_confirm / tc01_abort / tc01_status`
: TC-01 bench check test card.

`tc02_start / tc02_continue / tc02_abort / tc02_status`
: TC-02 captive hover.

`tc03_start / tc03_continue / tc03_abort / tc03_status`
: TC-03 truck fixed-wing.

`tc04_start / tc04_continue / tc04_abort / tc04_status`
: TC-04 first free flight.

`tc_export`
: Export test card results.

`tc_results`
: Show test card result summary.

Selected Telemetry Keys (v23)

`V_ms`
: Pitot airspeed (m/s)

`h_agl_m`
: Altitude AGL (m)

`vel_down_ms`
: Raw vertical velocity (m/s, negative=climbing)

- `vsi_smooth_fpm`
  5-beat smoothed VSI (ft/min) — use this for display
- `vsi_smooth_ms`
  5-beat smoothed VSI (m/s)
- `vsi_ref_fpm`
  Current VSI bug value (ft/min)
- `vsi_ref_err_fpm`
  VSI error from bug (ft/min)
- `ias_smooth_kt`
  5-beat smoothed IAS (kt)
- `hdg_ref_deg`
  Heading bug value (degrees)
- `hdg_ref_set`
  True if heading bug is set
- `hdg_ref_err_deg`
  Heading error from bug (degrees)
- `alt_ref_m`
  Altitude bug value (metres)
- `alt_ref_ft`
  Altitude bug value (feet)
- `alt_ref_set`
  True if altitude bug is set
- `alt_ref_err_ft`
  Altitude error from bug (feet)

`roll_rate_dps`
: Roll rate (deg/s)

`heading_deg`
: Current heading (degrees magnetic)

`motor_I_A`
: Motor current (A)

`bus_voltage`
: Battery voltage (V)

`wc_min_sep_m`
: Minimum wing tip separation all pairs (m)

`wc_clap_count`
: Validated clap events this session

`cic_mode`
: Climb intent: sustained/best_rate/best_angle/zoom

`mic_manoeuvre`
: Current aerobatic manoeuvre classified

`mic_urgency`
: Stick urgency scalar (0 to max)

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Green triangle (■) marks commands new in v23. For pilot/operator interface see MCO-1D Operator Manual v23.

Reference bugs (vsi_ref, hdg_ref, alt_ref) are display-only — the aircraft does not hold any reference value.